INTEGRATED SYSTEMS NEUROENGINEERING

UC San Diego
JACOBS SCHOOL OF ENGINEERING
Shu Chien-Gene Lay Department of Bioengineering

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TRACK 8 (A5L-09): *NEUROMORPHIC CIRCUITS & NEURON IMPLEMENTATIONS IMPLEMENTATIONS*

Adiabatic Energy-Recycling Charge-Redistribution Array

for ultra-low-power, massively parallel cross-correlation

Shashank Bansal · P. G. Hogganvik · S. Jain · G. Hota · B. Cherif · J. Leugering · G. Cauwenberghs

UC San Diego · U. Oslo · Qernel · Northrop Grumman · FZ Jülich

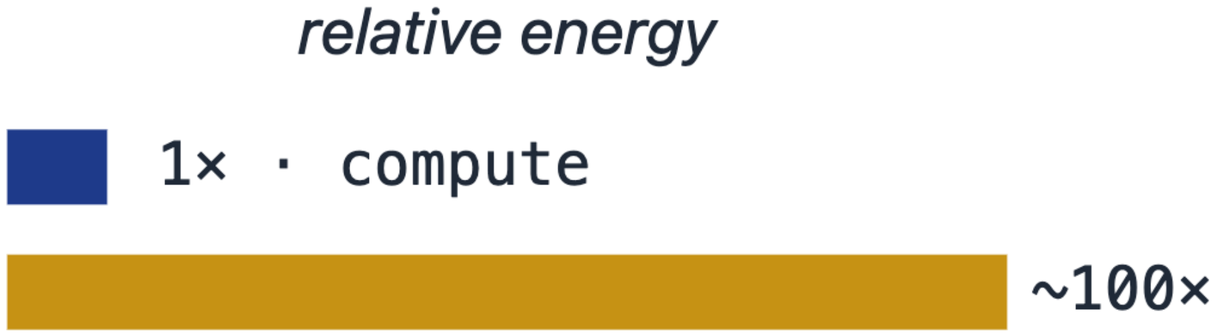
What’s the problem?

→Massively parallel cross-correlation
→must be ultra-fast, ultra-cheap

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The data-movement wall



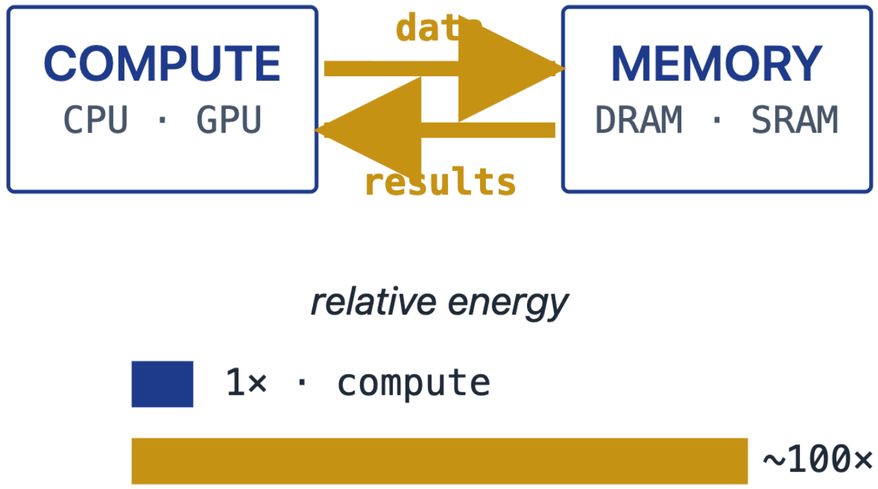


💡

moving data, not ~~compute~~, is the bottleneck for modern AI.

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The data-movement wall



The Adiabatic solution

01 **Compute where the data lives**

02 **Adiabatic recovery of all* energy**

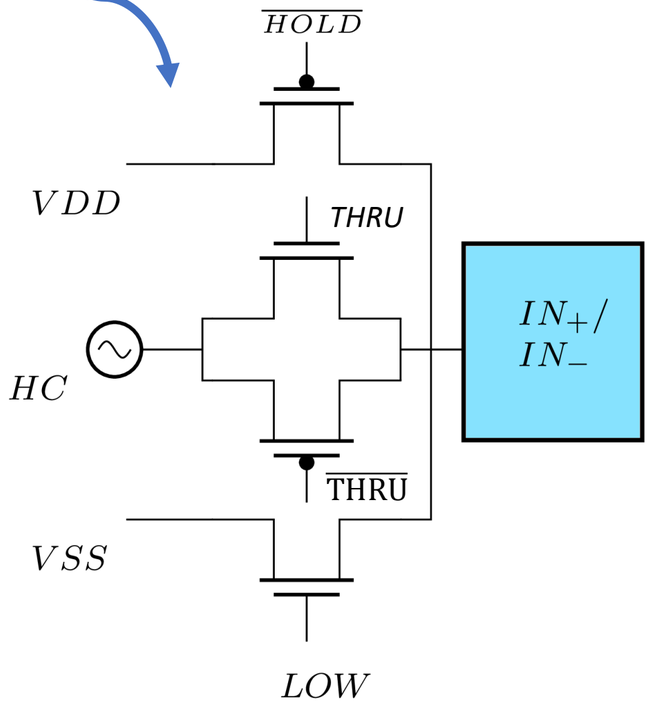
03 **Charge-domain accumulation**

See also: Kerneltron (Karakiewicz et al., 2003–2007)

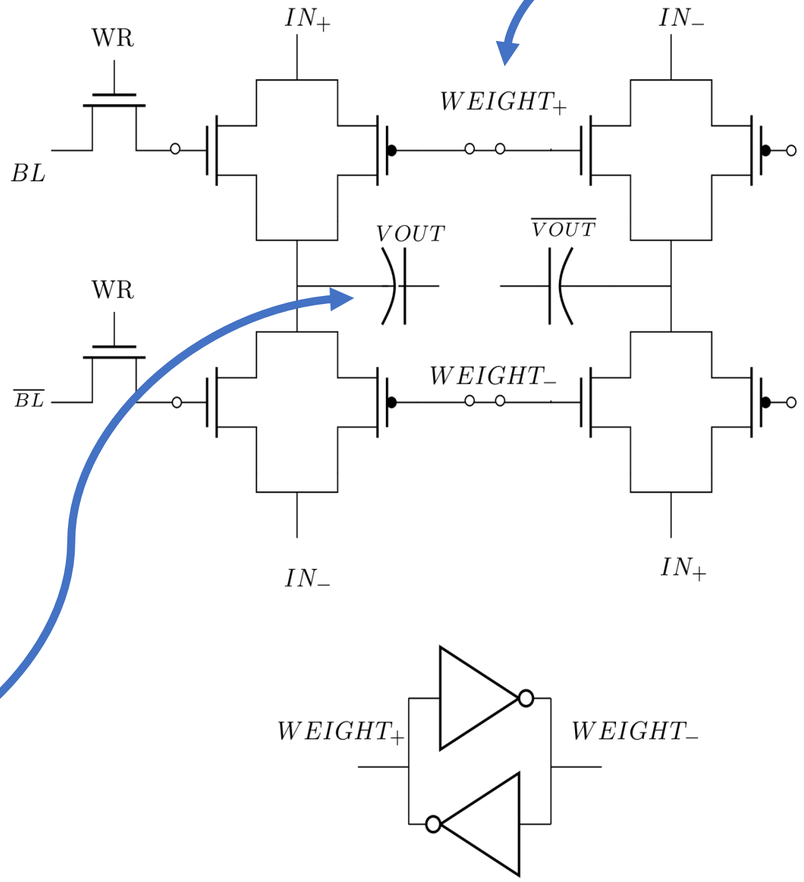
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Architecture at a glance

A diabatic input driver

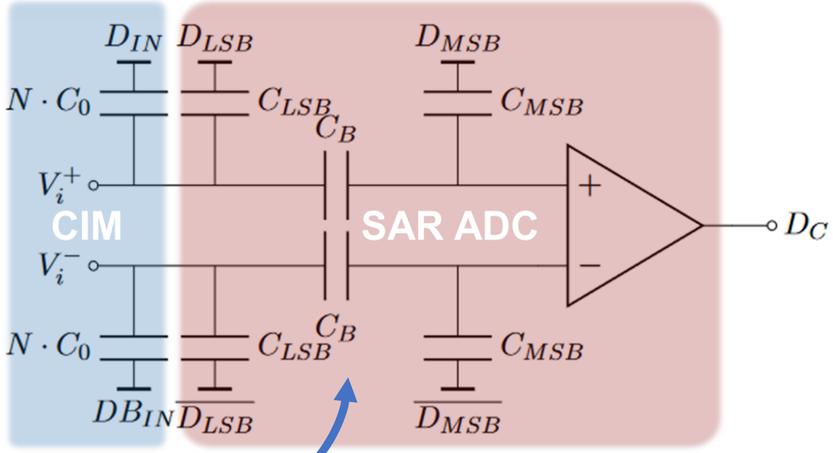


B ipolar 14T–2C SRAM cell

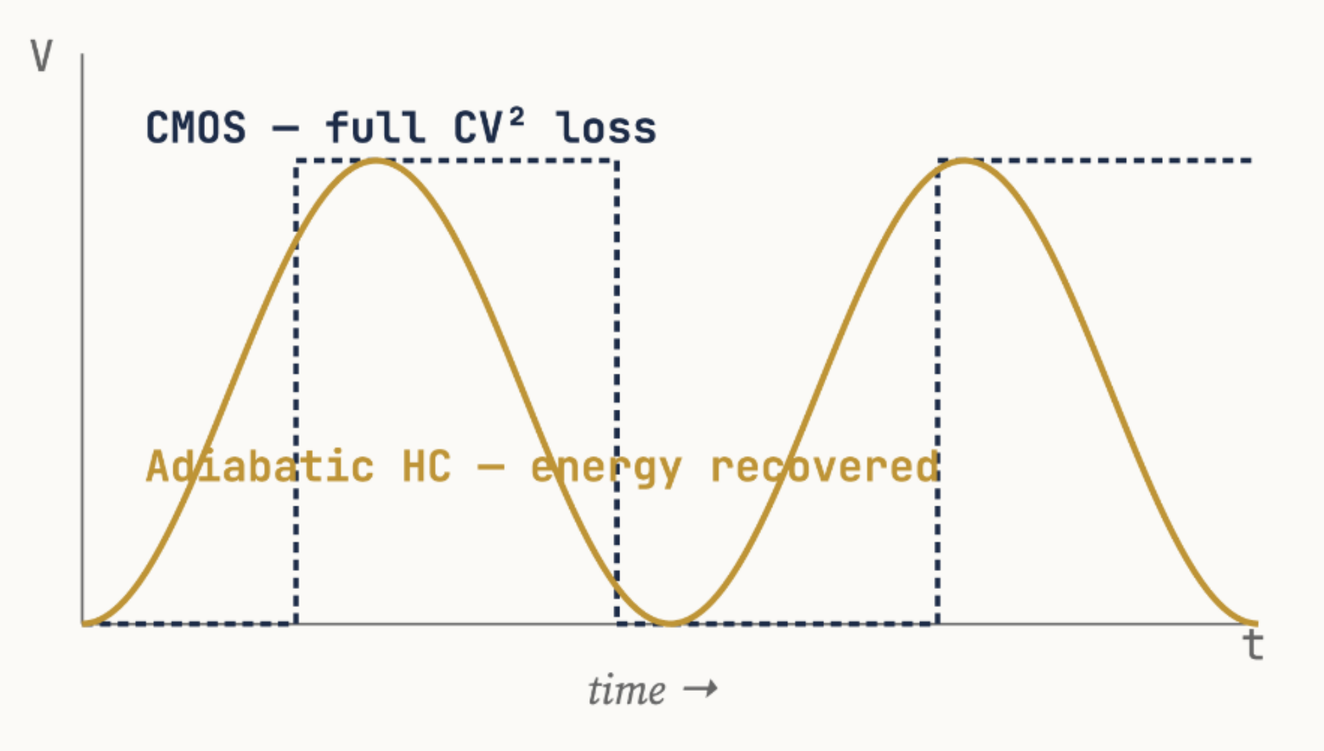


C harge-domain accumulation

D irect coupling to SAR ADC



A: Why *adiabatic* input drivers?



- sinusoidal “**hot clock**”

→ gradual charge / discharge

discharge

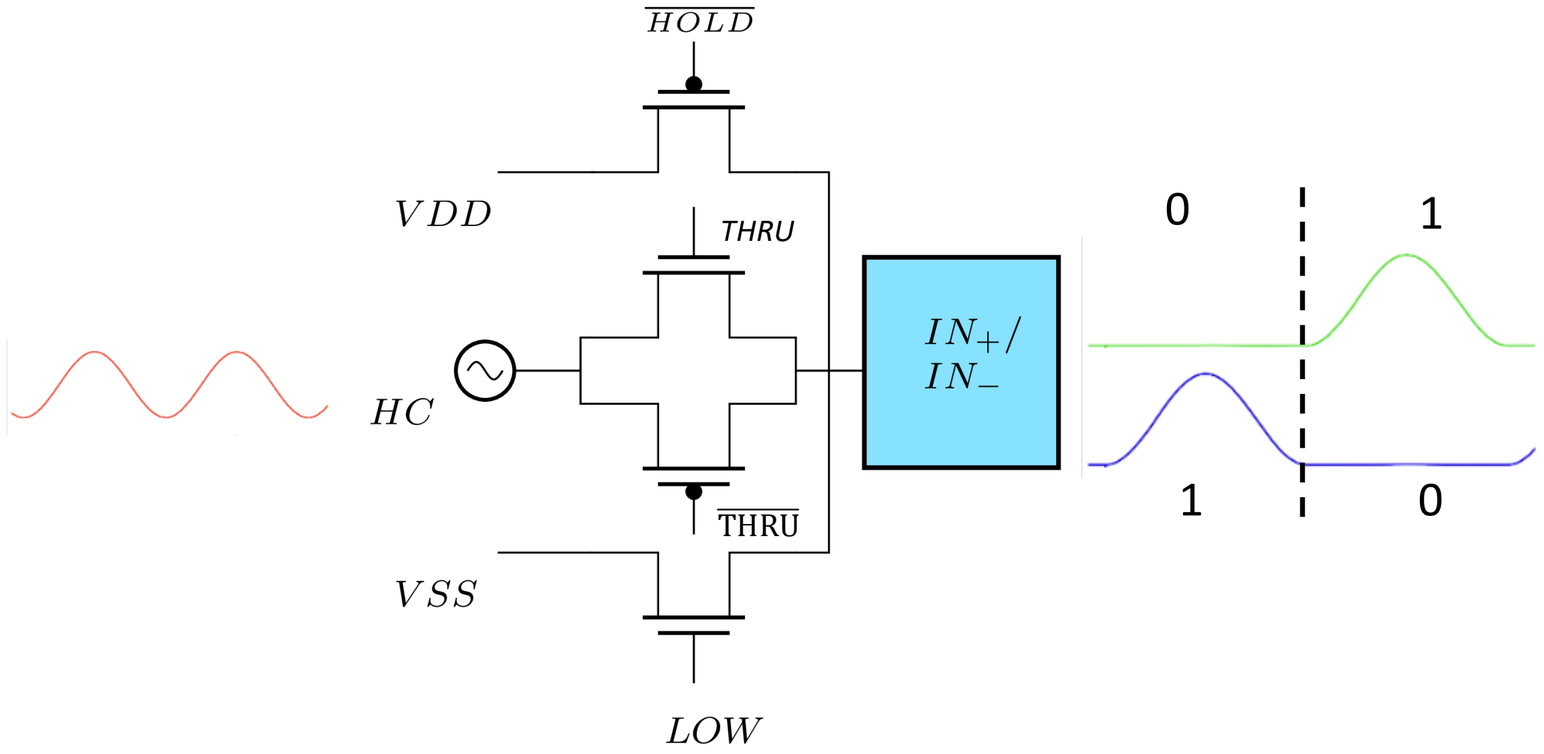
- only switch at zero voltage

voltage

- resonant tank recycles energy

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A: Adiabatic Input Drivers: Generation



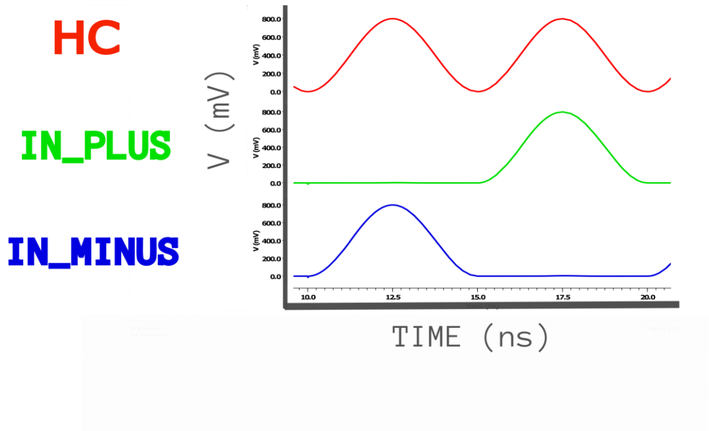


FIG. 2(B) · HOT CLOCK + DIFFERENTIAL OUTPUTS

bit = 1 → IN+ follows HC, IN– at GND
bit = 0 → IN– follows HC, IN+ at GND

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B: The 14T–2C unit cell

- 6T SRAM cell for bipolar weight
- symmetric coupling caps

→ XNOR-and-accumulate

- purely in charge domain
- Two phases: PROGRAM/COMPUTE

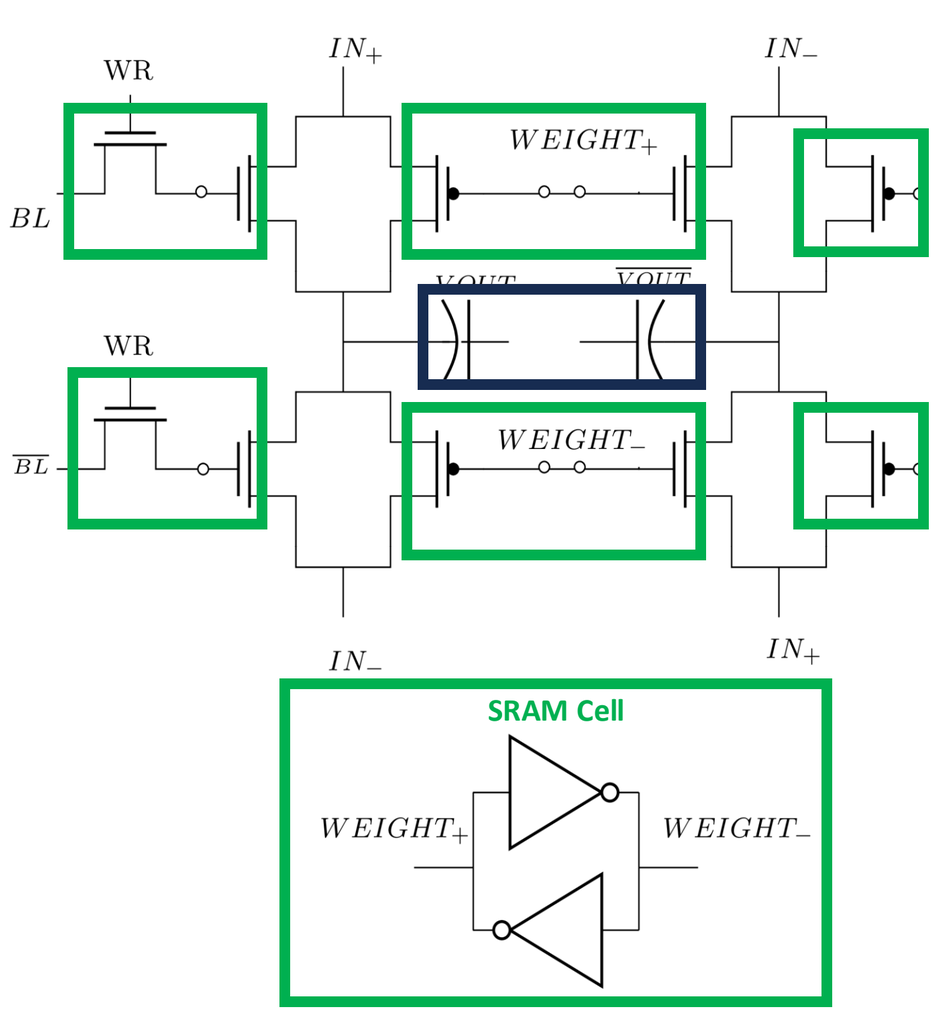
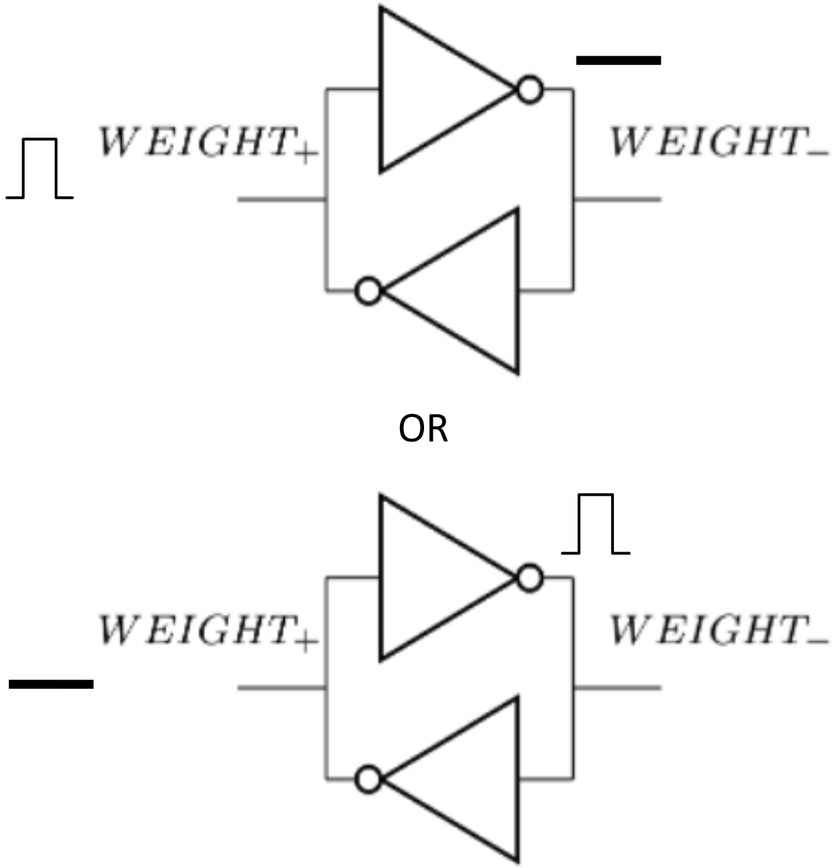


FIG 1: 14-2C SRAM UNIT CELL

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B: The 14T–2C unit cell

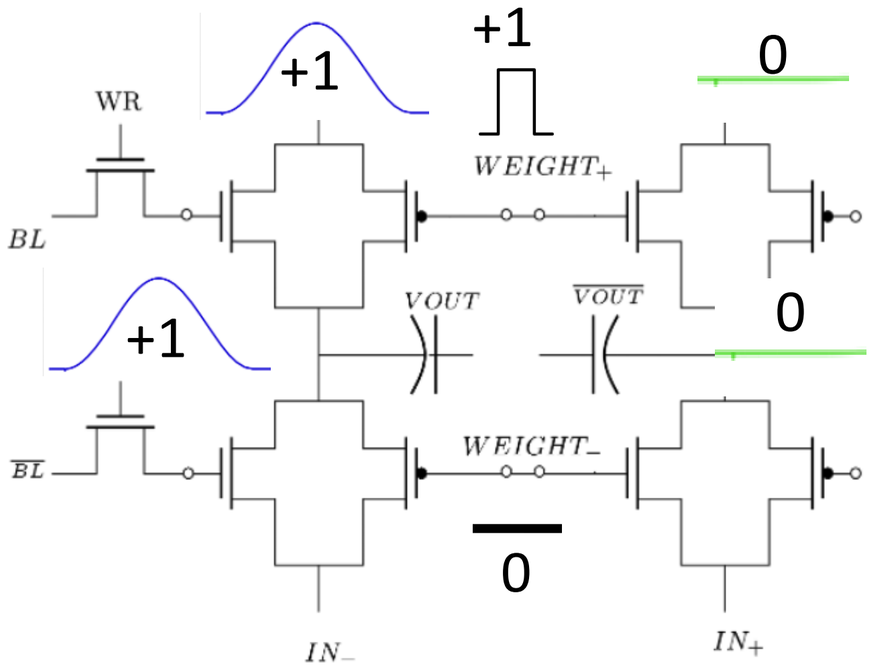


PROGRAM

Standard 6T SRAM-Cell for Binary Digital Weights

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B: The 14T–2C unit cell: COMPUTE phase



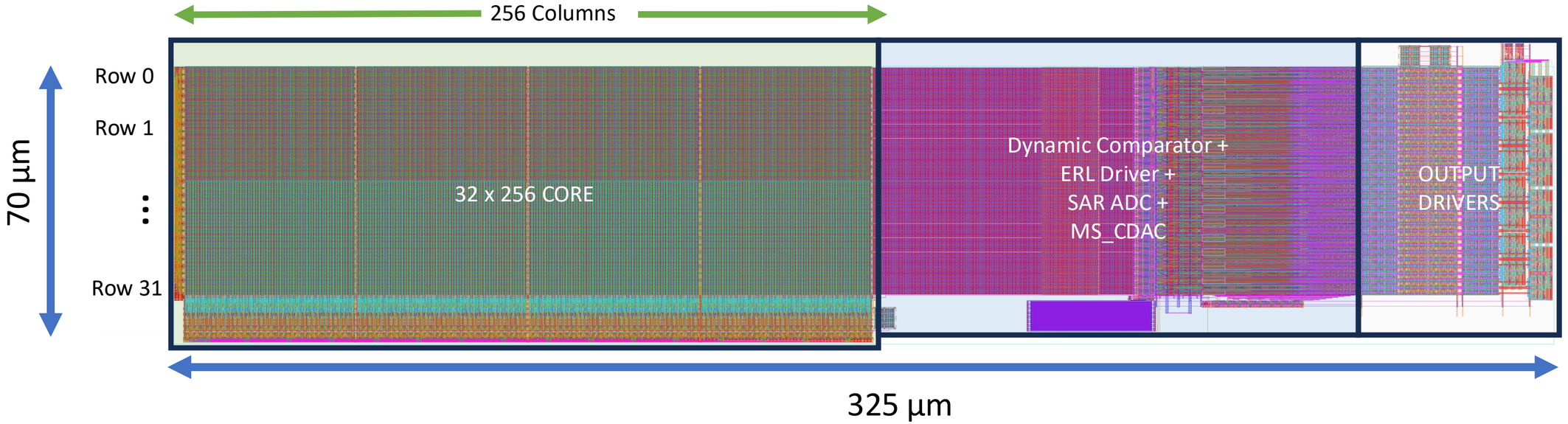
Step 1: Weights are stored in the SRAM during PROGRAM

Step 2: Adiabatic inputs generated during COMPUTE

Step 3: The circuit performs a signed XAC operation between the stored weights and column-parallel, complementary inputs that were adiabatically generated.

Step 4: After coupling through the CIM array, the voltage difference on the output nodes represents the accumulated correlation between the input and stored weight vectors

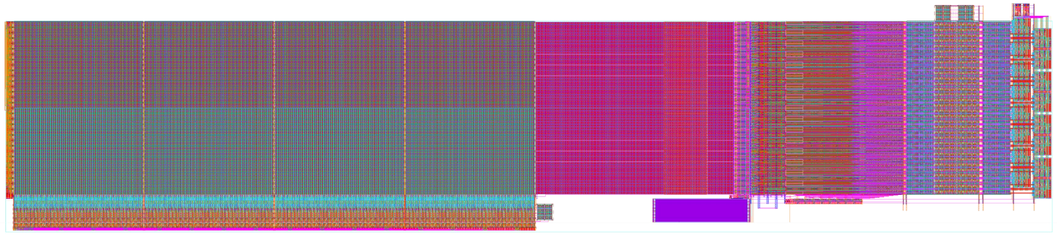
C: Charge Domain Accumulation: Full Array



- signed inner product weight × input
- computes row in single pass

$$\Delta V_{\mathrm{OUT}} \propto \frac{C_C}{C_{\mathrm{row}}} \sum_k \alpha_k \Delta V_{\mathrm{IN,k}}$$

C: Charge Domain Accumulation: Full Array



Each cell pushes charge onto a shared row line to form upto 257 unique voltages corresponding to 257 digital values

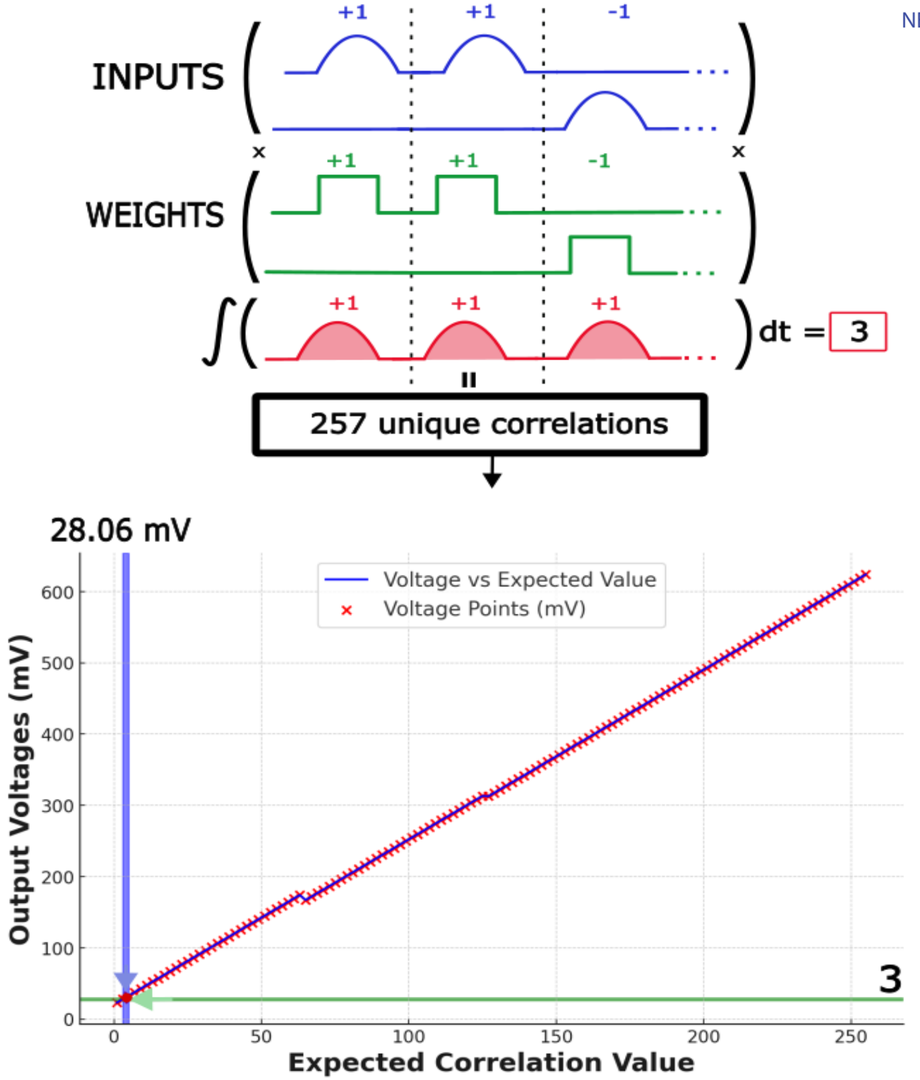
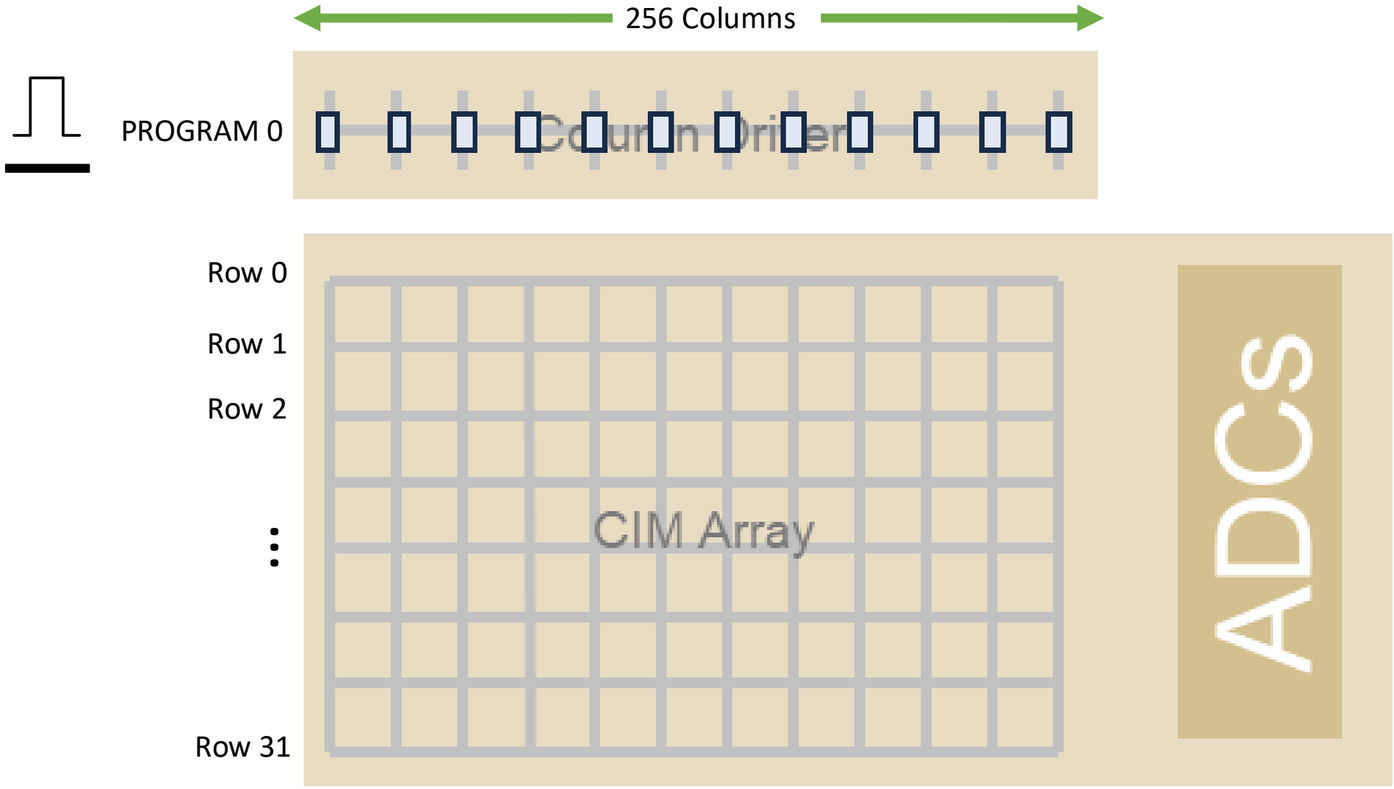


FIG. 3 • OUTPUT VOLTAGE VS. EXPECTED CORRELATION

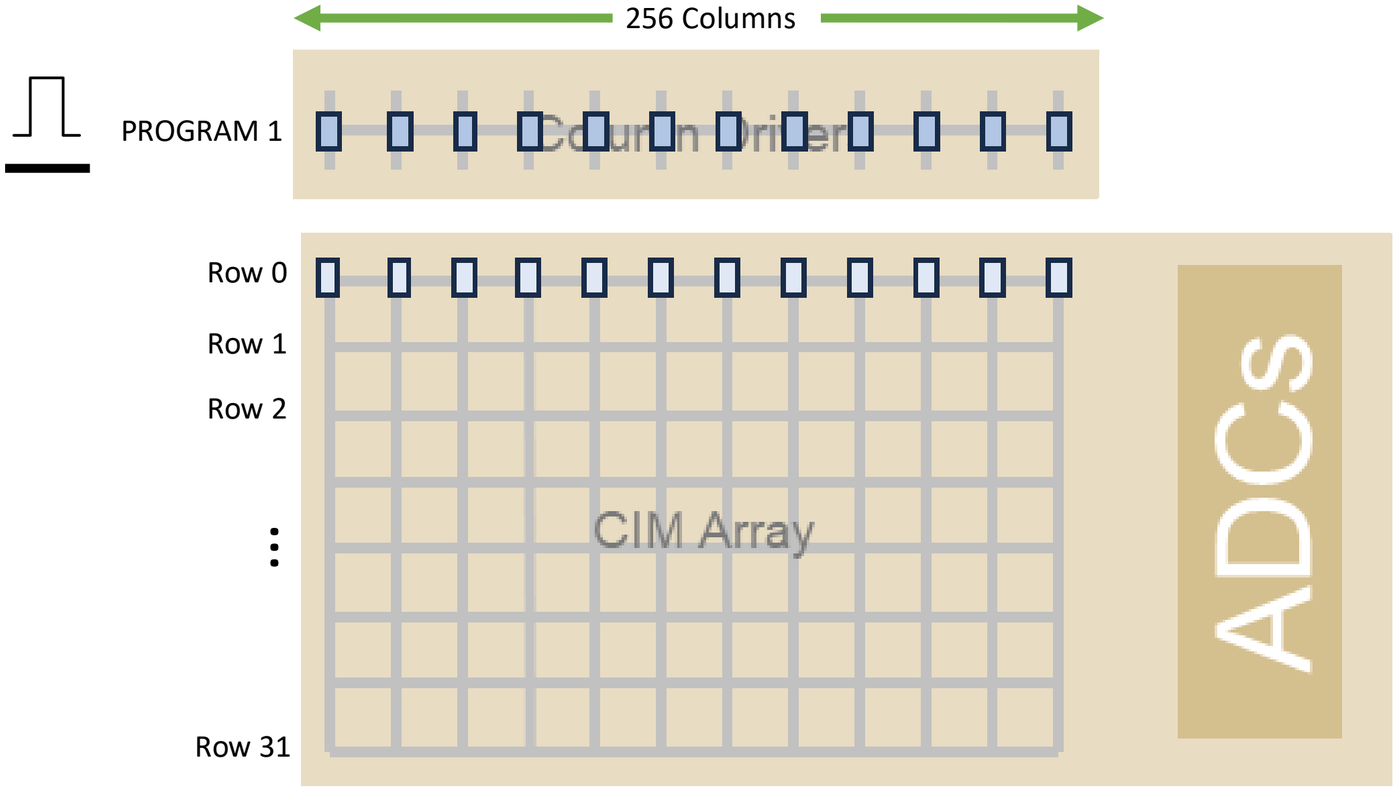
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PROGRAM: Full 32 x 256 CIM Array

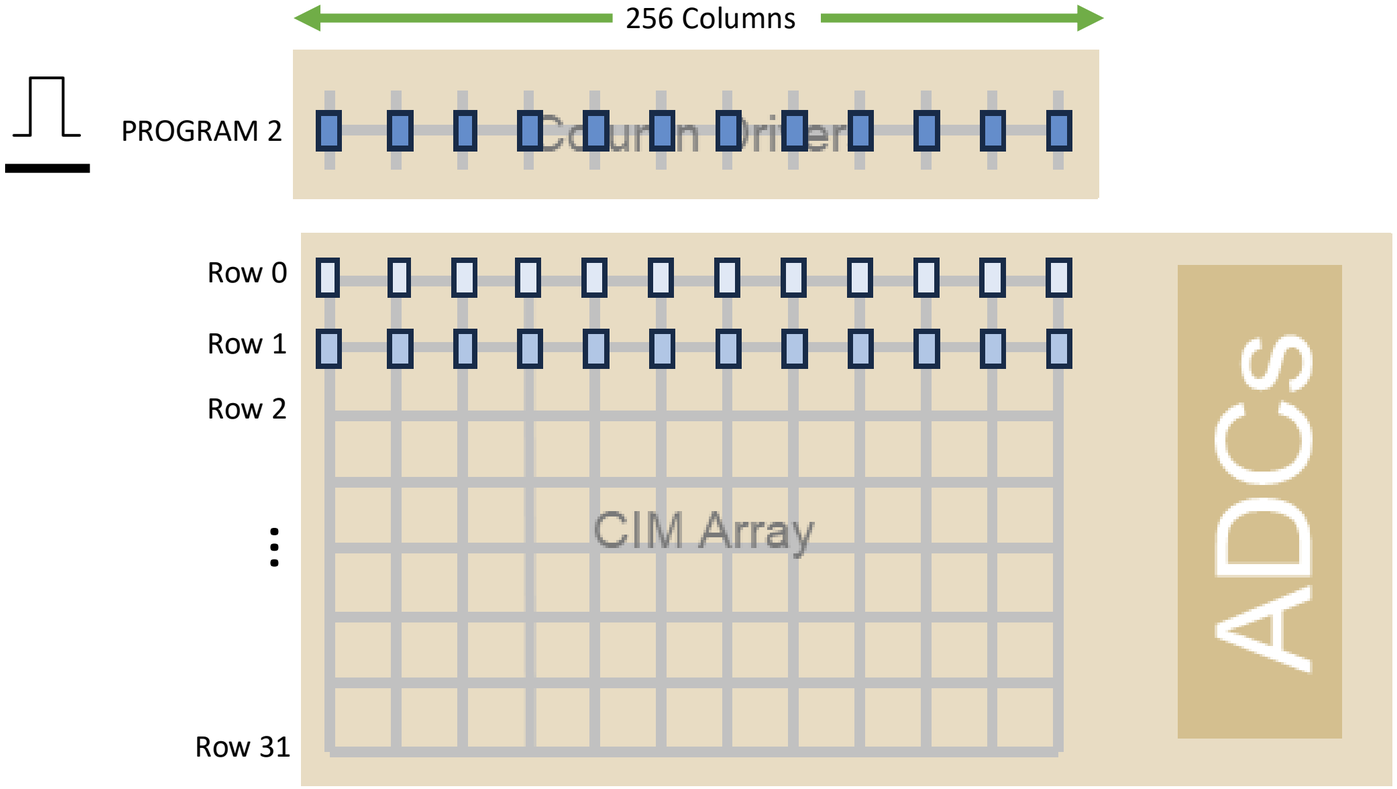


INTEGRATED SYSTEMS NEUROENGINEERING

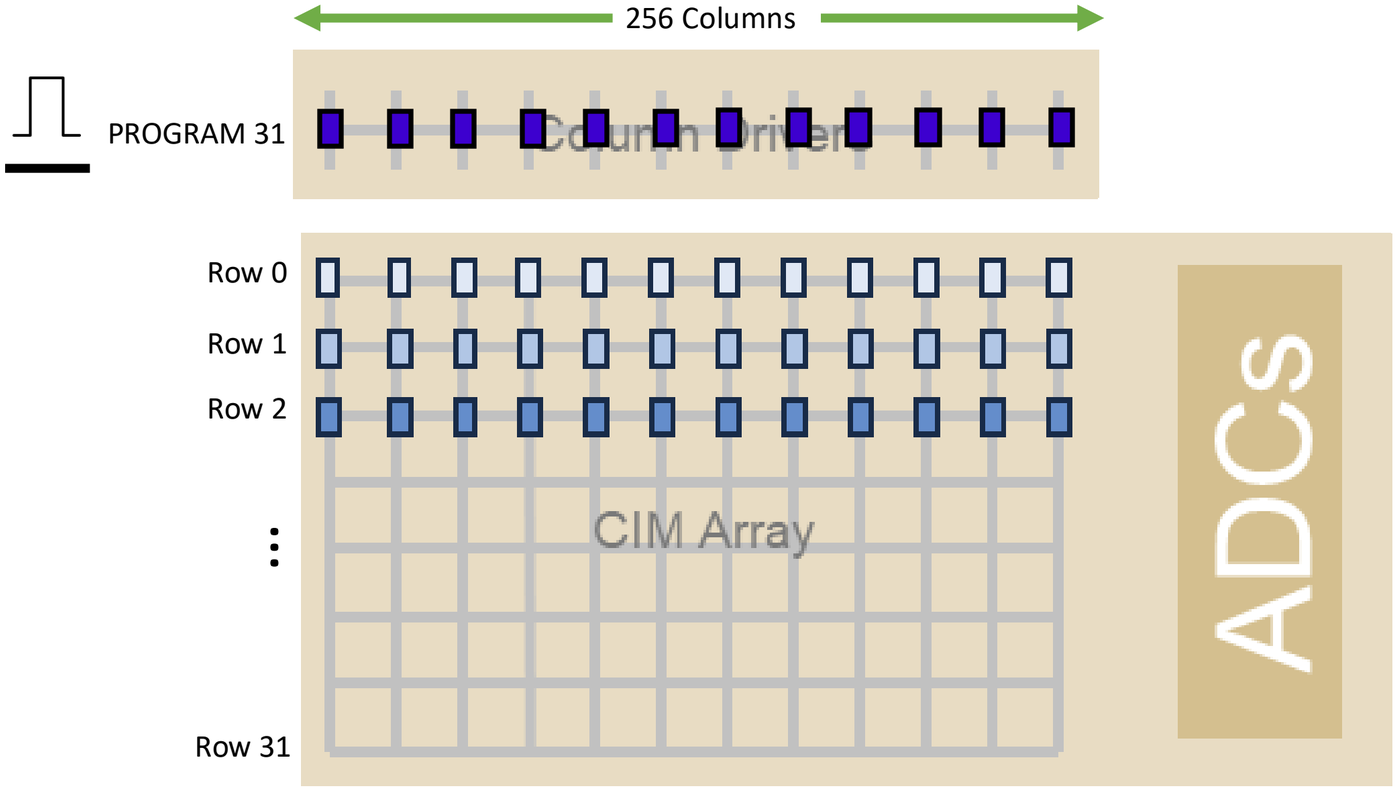
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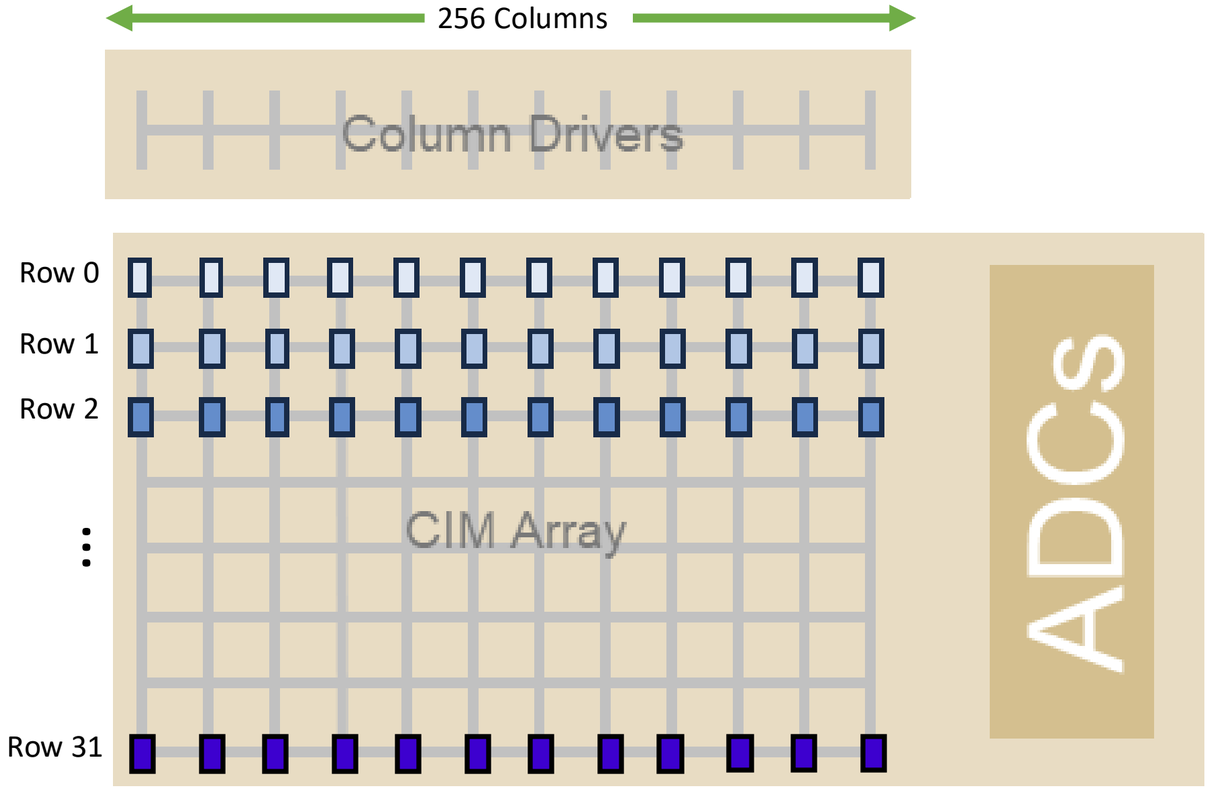
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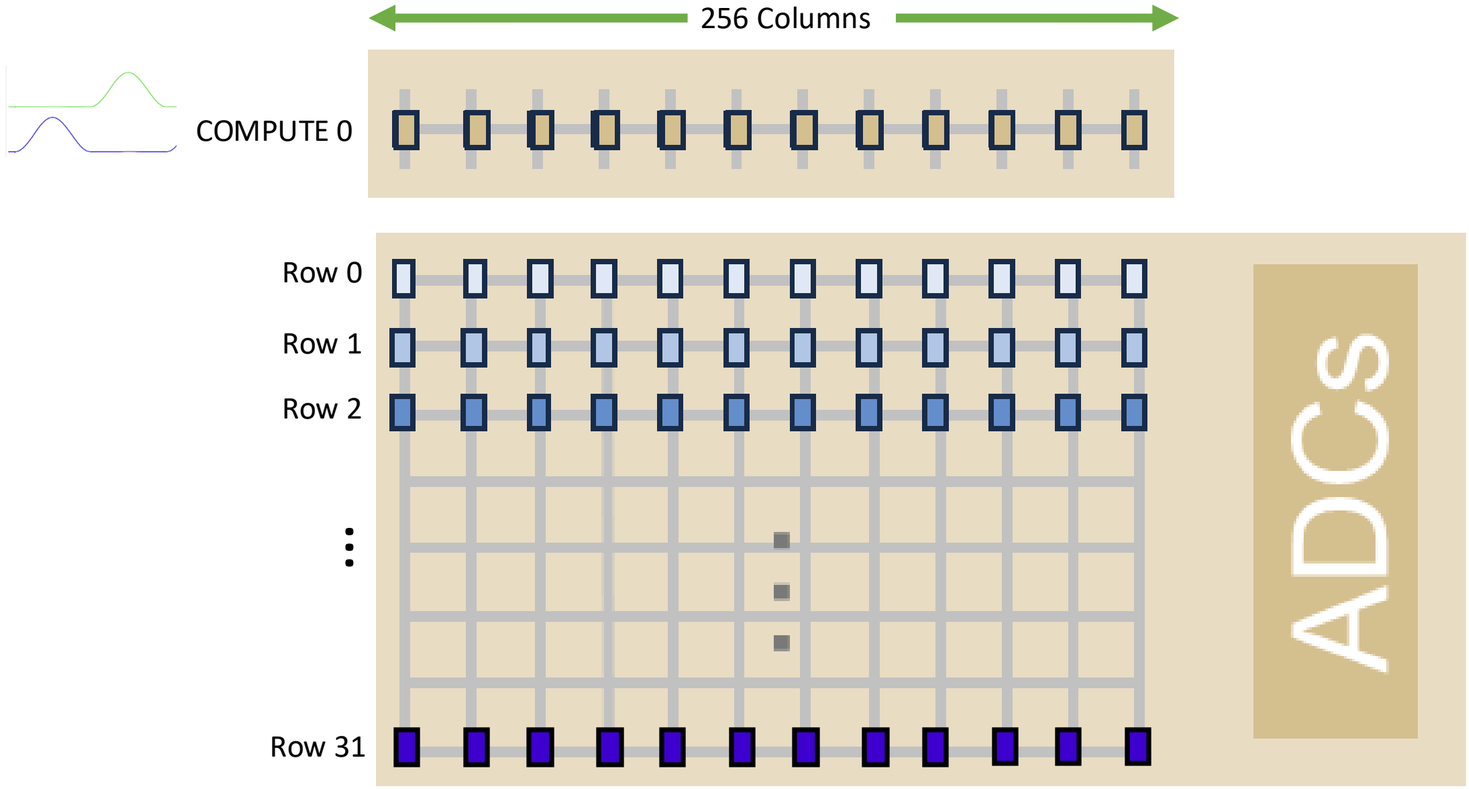
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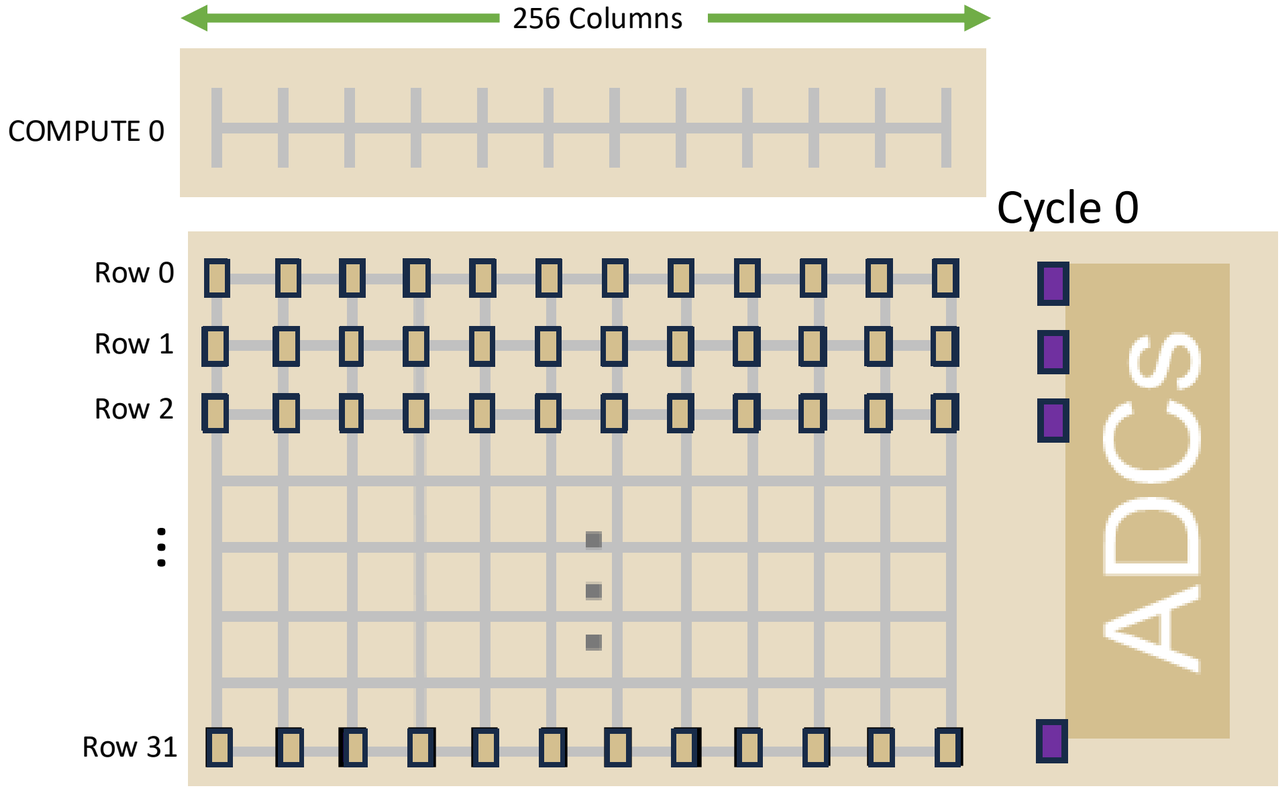
PROGRAM: Full 32 x 256 CIM Array



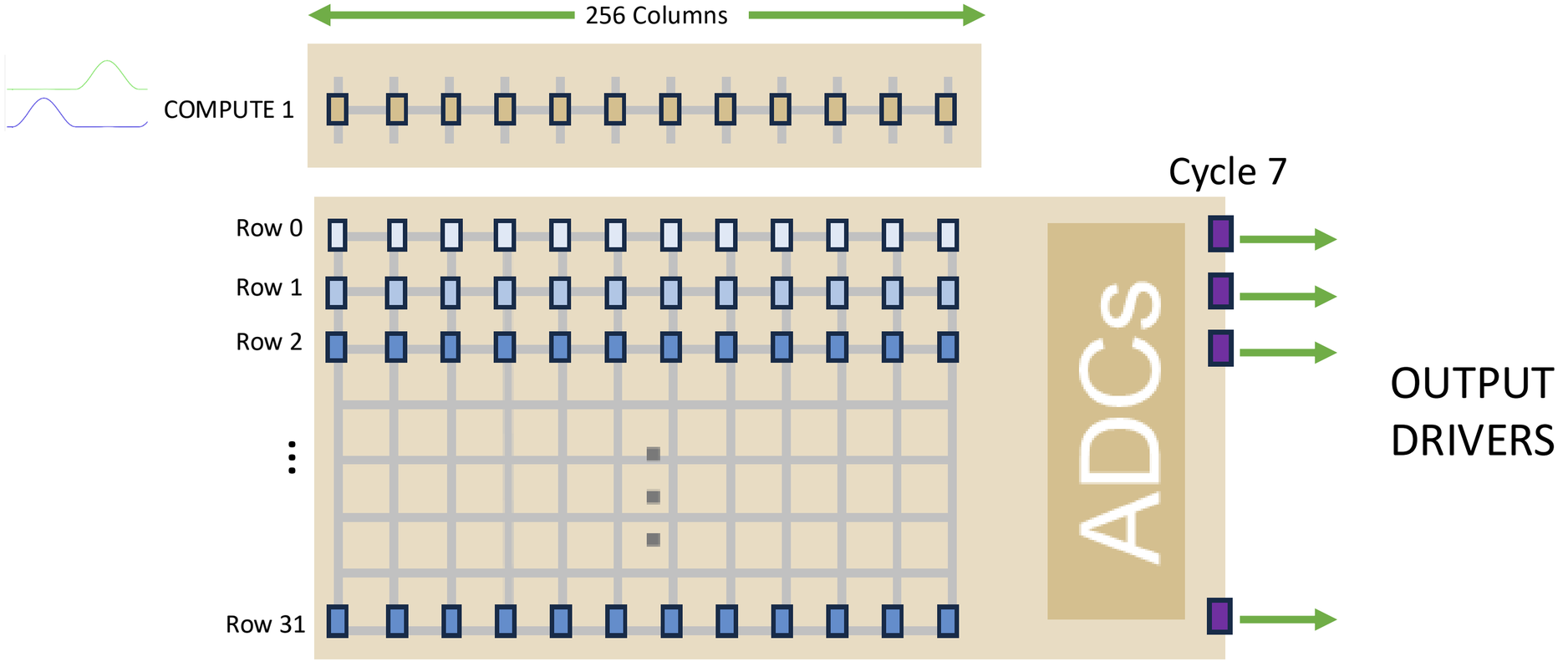
COMPUTE: Full 32 x 256 CIM Array



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COMPUTE: Full 32 x 256 CIM Array



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Where the energy goes

- Programming drives the $\mathrm{BL}/\overline{\mathrm{BL}}$ — a one-shot per weight current spike of *422 aJ/bit* however amortizes across thousands of input vectors.
- The cumulative-energy drops on every HC down-ramp (blue arrows).
- *Compute - **29 aJ/XAC op** as the overall steady-state cost.*

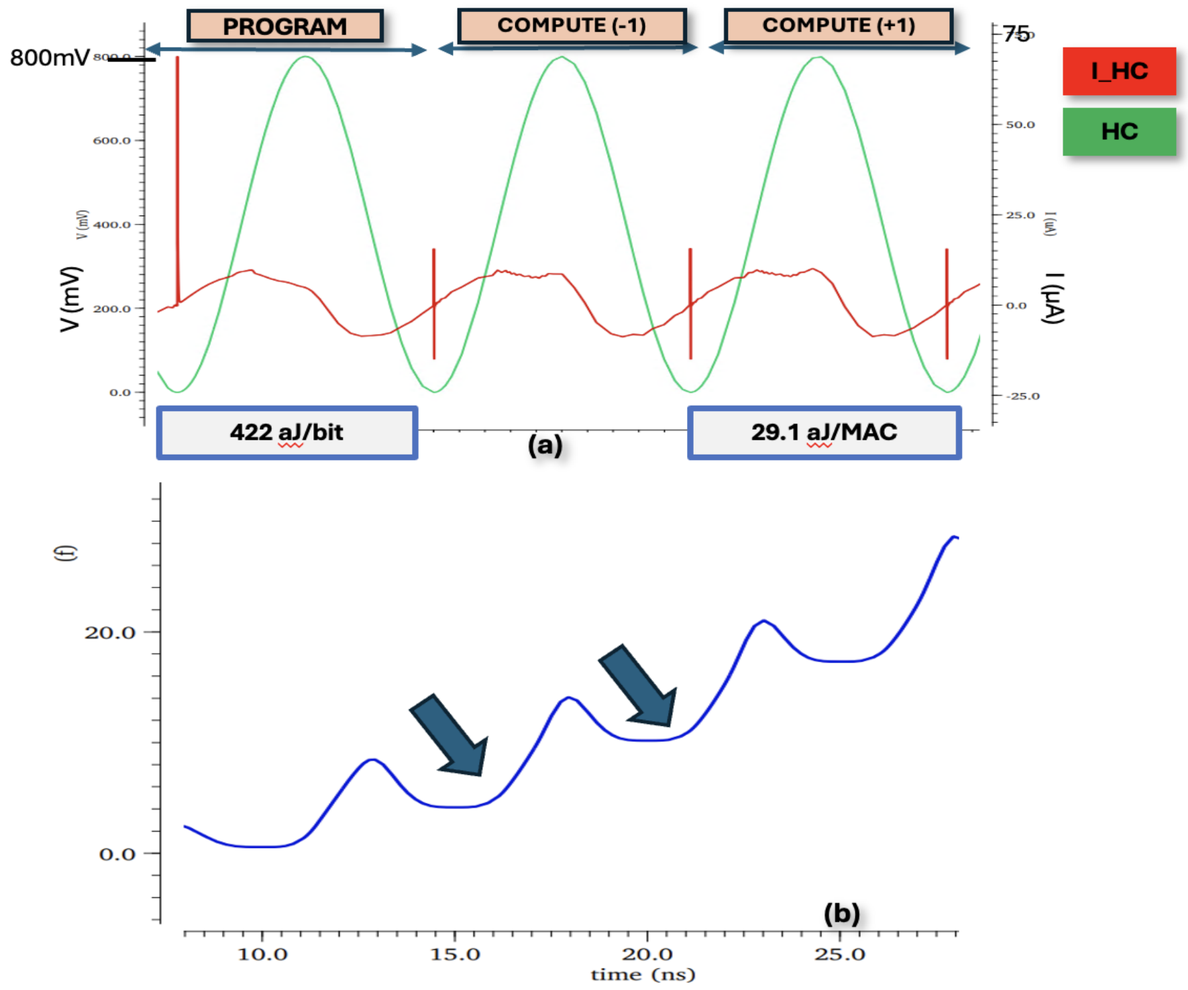


FIG. 4 • (A) HC + I_HC TRACE • (B) INTEGRATED ENERGY

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Getting to 29.1 aJ per XAC

1. Total row energy (8 compute cycles)
E_row, total = 60.29 fJ − 0.563 fJ = ***59.73 fJ***

2. Per-cycle row energy
E_row,cycle = 59.73 / 8 = ***7.46 fJ***

3. XAC per cell per cycle
E_cell,XAC = 7.46 fJ / 256 cells = **29.1aJ / XAC**

Energy efficiency:

- ✓ post-layout simulation analysis
- ✓ **1 × 256** row @ **200 MHz**

The 0.563 fJ subtraction is the leakage floor measured during a no-compute window — everything above that line is real switching + recovery activity.

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>1500 TOPS/W

W

Energy efficiency:

✓ post-layout simulation analysis

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Performance comparison among recent SRAM-CIMs

Over 30× lower XAC energy than any prior SRAM-CIM, across several process nodes — including advanced 7 nm and 16 nm designs.

REFERENCE	VENUE	NODE	SUPPLY (V)	EFFICIENCY (TOPS/W)
Sinangil et al., '21	JSSC	7 nm	0.8	351
Su et al., '21	ISSCC	28 nm	0.8	22.75
Jia et al., '22	JSSC	16 nm	0.8	30
Zang et al., '23	ISCAS	40 nm	0.7–0.9	282–607
Karakiewicz et al., '07 *Kerneltron — charge-injection, historical*	JSSC	350 nm	1.65	960
This work	—	**22 nm**	**0.7–0.9**	**>1500**

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A fit for neuromorphic & neural interface computing

01 Correlation is the primitive

Spike-train cross-correlation, template matching, and STDP learning all reduce to bipolar inner products — exactly what this array computes.

02 Scalable to closed-loop control

- The array's MVM throughput allows for real-time decoding of multi-channel neural recordings.
- Enables on-implant classification and closed-loop stimulation without off-chip data transfer.

03 Energy headroom for always-on, event-driven operation

- Sub-aJ / XAC makes continuous, parallel sensing practical within the strict 15–40* mW thermal budget of chronic intracortical implants.
- Adiabatic energy recovery activates the array only when it draws power during compute

** Rapeaux & Constandinou, 2021. Frontiers in Neuroscience.*

Acknowledgements

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Bouchaib Cherif (NGC)





